Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years of experience building interactive data platforms, API-backed web applications, and geospatial-style decision workflows through structured datasets, backend integration, testing, and stakeholder-facing documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built data-backed web platform components across **Azure SQL, PostgreSQL, and cloud storage** for 500+ users, improving dashboard response from **1.2s to 350ms** through schema and query optimization.
- Implemented validation/release workflows for 5K+ monthly jobs with reconciliation checks, improving data quality and long-term maintainability for decision-support interfaces.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed a **Flask + React** interactive platform handling 10K+ daily requests with modular APIs, secure ingestion, and user-facing flows that translated requirements into functional interfaces.
- Collaborated with product/design stakeholders on API and UI refinements; added CI/CD and operational checks to improve reliability and reduce turnaround time by 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered integration pipelines and API workflows with **AWS Glue/S3**, OOP patterns, and testing automation, improving response times by 30% and pipeline efficiency by 68%.
- Prepared and validated cross-platform datasets/workflows with Jenkins, PyTest, and Mockito across 3 OS environments, strengthening traceability and reproducible issue triage.

Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built a MERN operations platform with REST APIs and WebSockets supporting 8+ workflows and 2M+ synchronized records, with clear metadata and service contracts for maintainable expansion.
- Automated deployment/testing with Jenkins, Docker, and Azure SQL services, reducing deployment time by 50% while improving consistency of production releases.

LST Capitals **India**
Web Developer Intern *November 2022 - January 2023*
- Developed cross-platform analytics flows with Flask API and Firebase for 50K+ records, supporting interactive views and near real-time operational summaries.
- Built backend integrations and documented webhook-driven automations for billing/payment workflows, improving supportability and partner onboarding speed.

Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, C++
Web & Mapping Stack: React.js, HTML, CSS, Flask, Node.js, REST APIs, JSON, GeoJSON familiarity
Geospatial/Data: Raster/Vector data workflows, projection-aware data handling, metadata documentation, data quality validation
Backend & Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Firebase
Cloud & Tooling: AWS (S3, Glue, RDS), Docker, Kubernetes, Git, Jenkins, GitHub Actions, CI/CD
Engineering Practices: Testing, debugging, performance tuning, cross-functional requirement translation, technical documentation

Projects

HMIS – Hospital Management Information System Live | *MERN, ABDM APIs, Twilio*
- Designed an interactive operations interface with structured API integrations and monitoring-backed workflows, supporting **100+ daily appointments** and improving record retrieval by **2x**.

FinNexus Live | *Next.js, TypeScript, MongoDB*
- Built map-style analytics-ready transaction views and secure workflow controls for **50K+ users** and **1M+ transactions**, with scalable deployment and real-time updates.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built a data exploration workspace spanning **10+ databases** with schema-aware querying and structured summaries, enabling faster analysis and decision-support reporting.